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METADATA REPORT

DATA IDENTIFICATION:	
Title	EstuarineEcosystemMap2018_19092019
Description (detailed)	This is the most recent version of the National Estuary Ecosystem Map. The map was developed as a collaborative effort and delineates the Estuarine Functional Zone for South Africa (Van Niekerk et al. 2019a). Each EFZ was then assigned an ecosystem type according to Van Niekerk et al (2019b). In South Africa, the EFZ is defined as the area that not only encapsulates the estuary waterbody, but also the supporting physical and biological processes necessary for estuarine function and health. It includes all dynamic areas influenced by long-term estuarine sedimentary processes, i.e. sediment stored or eroded during floods, changes in channel configuration, aeolian transport processes, and/or changes due to coastal storms. It also encompasses the multiple ecotones of floodplain and estuarine vegetation that contribute detritus (food source) to the estuary and/or provide refuge during high flow events. The EFZ captures the natural, historical estuarine extent and should not be confused with setback/management lines that often exclude developed areas. The EFZ purpose is to identify the 'space' in which estuarine physical and biological functions take place over long time scales (>decades). Development in the EFZ is captured as an aspect of habitat degradation or decline in overall estuary condition. The upstream boundary of the estuaries was determined as the limits of tidal variation or salinity penetration, whichever penetrates furthest. The estuary mouth was taken as the downstream boundary of an estuary. The highly dynamic nature of this area presents a significant challenge to accessing change in biodiversity and even managing estuaries. To account for this, and to allow for a seamless integration with the Marine and Coastal Realm, the concept of 'Estuarine Shore' was introduced to reflect the dynamic nature of the interface between estuaries and the coast. Estuarine Shores refers to sand berms or bars that form in front of estuaries. They vary substantially in size and shape over decadal scales and can be completely absent during a flood or a near permanent feature during periods of low flow. Estuarine Shore was defined as the area from the base of the foredune, or where this line would be if dunes were present, to the back of the surf zone. The full extent of the Estuarine Shores is encapsulated in the EFZs and not be considered separate from the functional unit. The surfzone was included to reflect a continuum in estuarine-marine connectivity through estuarine inputs to the surfzone, either as direct flow through an open mouth or in seepage through the berm in a closed system In total 290 estuaries and 42 micro-estuaries were delineated and classified into 22 estuary ecosystem types and 9 micro-system types. The estuary classification is based on 4 biogeographical regions (Cool Temperate, Warm Temperate, Subtropical and Tropical) and 9 estuary categories (Estuarine Lake: Estuarine Bay, Estuarine Lagoon, Predominantly Open, Large Temporarily Closed, Small Temporarily Closed, Large Fluvially Dominated, Small Fluvially Dominated, Arid Predominantly Closed). While the micro-estuary ecosystem classification is derived from one micro-system type across 3 biogeographical region.
Purpose	To delineate the physical and biological processes and habitats necessary for estuarine function and health based on their historical extent and Estuarine Ecosystem Types at the landscape scale.

Status	Estuary Ecosystem Map 2018, version used in the 2018 National Biodiversity Assessment
Maintenance and update frequency	This version will be updated periodically as new data emerges.
Topic category	Ecosystem Classification and mapping
Lineage	This version of the estuarine ecosystem map is an improvement on previous versions that were based on the 1:10 000 scale topographical maps 5m contour. The Estuary Classification is an update of the Whitfield (1992) classification scheme. This version incorporates biogeographical zonation and introduces new types.
Citation	Van Niekerk, L., Adams, J.B., Fernandes, M., Harris, L., Lamberth, S.J., MacKay, CF., Petersen, C., Ramjukhad, C.-L., Riddin, T., Van Deventer, H. & Weerts, S.P., 2019, ' Chapter 4: Extending the Estuary Functional Zone to include key habitats and processes', South African National Biodiversity Assessment 2018: Technical Report. Volume 3: Estuarine Realm. South African National Biodiversity Institute, Pretoria. Report Number: SANBI/NAT/NBA2018/2019/Vol3/A
Keywords	Estuary, estuarine functional zone, map, classification, ecosystem type.

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Role	Lara van Niekerk is the scientific co-ordinator of data collected for the National Estuary Map and assessment.
Funders	Council of Scientific and Industrial Research and South African National Biodiversity Institute

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Role	Author; Custodian; Distributor; Owner; Point of contact; Processor; User.

RESOURCE CONSTRAINTS:	
Access constraints	Copyright holder: South African National Biodiversity Institute
Use constraints	This data may not be reproduced by any means, nor redistributed via web site or ftp site, without prior permission. Whilst every effort has been made to ensure the accuracy of data, users are advised to use the data and conclusions drawn from its use with caution. Users noting errors and omissions are requested to notify Lara van Niekerk to improve data accuracy. These data are not for resale or replicating. This digital version is in the public domain requiring only the conventional acknowledgement of source in publications and reports.
Other restrictions	APPROPRIATE USE OF THIS DATASET: The Estuary Ecosystem Map has been created for national purposes and the boundaries are mapped at a 1:10 000 scale. More information on the clarity around appropriate use of the data can be discussed by contacting us.

SPATIAL RESOLUTION:	
Spatial representation type	Geodatabase, Text, Vector.
Equivalent scale - Denominator	Various (See technical documents) 1:3000 to 1:50 000

COORDINATE REFERENCE SYSTEM: Custom	
Projected/Geographic coordinate system	PROJCS["AEA_RSA_WGS84",GEOGCS["GCS_WGS_1984",DATUM["D_WGS_1984",SPHEROID["WGS_1984",6378137.0,298.257223563]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433]],PROJECTION["Albers"],PARAMETER["False_Easting",0.0],PARAMETER["False_Northing",0.0],PARAMETER["Central_Meridian",25.0],PARAMETER["Standard_Parallel_1",-24.0],PARAMETER["Standard_Parallel_2",-33.0],PARAMETER["Latitude_Of_Origin",0.0],UNIT["Meter",1.0]]

TEMPORAL EXTENT (time period covered by the content of the dataset):	
Temporal extent - Begin date	2014
Temporal extent - End date	2018

ONLINE RESOURCE (additional resources available online):	
Description	The National Estuary Map includes the delineation of the Estuarine Functional Zone and Estuary Ecosystem Classification. However, for information about the estuarine vegetation, please refer to the NMU Estuarine Vegetation Database.
Linkage	

LEGEND PROPERTIES:

Classification	See attribute fields

DETAILED NOTES:	
Supplemental information	Van Niekerk, L., Adams, J.B., Fernandes, M., Harris, L., Lamberth, S.J., MacKay, CF., Petersen, C., Ramjukhad, C.-L., Riddin, T., Van Deventer, H. & Weerts, S.P., 2019a, ' Chapter 4: Extending the Estuary Functional Zone to include key habitats and processes', South African National Biodiversity Assessment 2018: Technical Report. Volume 3: Estuarine Realm. South African National Biodiversity Institute, Pretoria. Report Number: SANBI/NAT/NBA2018/2019/Vol3/A Van Niekerk, L., Adams, J.B., James, N., Lamberth, S.J., MacKay, C.F., Turpie, J.K., Rajkaran, A., Weerts, S.P. & Whitfield, A.K., 2019b, 'Chapter 3: A new Ecosystem Classification for South African estuaries' in South African National Biodiversity Assessment 2018: Technical Report. Volume 3: Estuarine Realm. South African National Biodiversity Institute, Pretoria. Report Number: SANBI/NAT/NBA2018/2019/Vol3/A

The attribute field data can also be provided as a MS Excel spreadsheet and uploaded as an online resource.

ATTRIBUTE FIELDS

Field Name	Full name	Description
OBJECTID	Object Identification	Internal feature number.
Shape	Shape	Feature geometry.
EstuaryID	Estuary identification	Unique code of individual estuaries in 2018
Estuary_Name	Map code 2018	Name of individual estuaries in 2018
EcosystemType_Primary	Name of Primary Ecosystem Type	Name of the Estuarine Ecosystem Type.
BroadGroup	Broad Group	Name of the Broad Group indicating whether an ecosystem is classified as an Estuary or a type of Micro-System (in this layer only micro-estuaries are mapped)
Bioregion	Bioregions	This is a level in the classification system that describes the region in which the ecosystem type is nested.
EFZ	Estuarine Functional Zone	Indicated that the polygon is part of the Estuarine Functional Zone.
PolygonColour_a	Polygon Colour	This is the field that the layer file uses to create the colours for each ecosystem
Crossrealm_OtherMaps	Cross realm fields from other ecosystem maps	This field indicates where a polygon is also represented in another ecosystem map (i.e. terrestrial, marine, coastal). In some cases only a portion of the map is represented by the in another ecosystem map.
EcosystemType_Secondary	Ecosystem Type Secondary	This is the name of the polygon as it is classified in other realm maps.
Shape_Length	Shape Length	Automatically generated geodatabase value indicating the length of the polygon in meters
Shape_Area	Shape Area	Automatically generated geodatabase value indicating the length of the polygon in square meters

The descriptions of the metadata fields are provided and copyright of Open Source Geospatial Foundation. Available at https://geonetwork-opensource.org/manuals/2.10.4/eng/users/appendix/glossary_of_metadata/index.html